

Topological Foundations: Jordan Curve Theorem

Formalization Notes

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Definition 1 (Homotopy and Nullhomotopy). *If f and f' are continuous maps of the space X into the space Y , we say that f is **homotopic** to f' if there is a continuous map $F : X \times I \rightarrow Y$ such that*

$$F(x, 0) = f(x) \quad \text{and} \quad F(x, 1) = f'(x)$$

*for each $x \in X$. (Here $I = [0, 1]$.) The map F is called a **homotopy** between f and f' .*

*If f is homotopic to f' , we write $f \simeq f'$. If $f \simeq f'$ and f' is a constant map, we say that f is **nullhomotopic**.*